Analytics Expansion for the Listen Up Collaborative: Review of Existing Tools and Future Directions

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INTRODUCTION

On Thursday, December 4, 2025, I had an initial meeting with Adrian Wolf (Great Peninsula Conservancy), and Carrie Clendaniel (Jefferson Land Trust), with the Listen Up Collaborative. They explained their effort to improve land management by tracking species recovery in treated areas by tracking sound duration and frequency of target/indicator species' sounds in treatment and control sites.

The Listen Up Collaborative currently uses AudioMoth devices for audio data collection and Arbimon for data organization and pattern matching. Adrian and Carrie expressed two key pain points with the data current process:

- Data management issues with the existing 400,000 recordings, and concerns about adding new species and records to the dataset.
- The overwhelming manual effort required for the validation process, post-pattern matching.

RESEARCH QUESTIONS

1. Understanding the full capacity of Arbimon
    a. Currently only the pattern matching analysis tool is being utilized to detect presence/absence. What other audio analysis tools could be of use?
        i. i.e. clustering, random forest, sound scape analysis, audio event detection
2. How to improve data efficiency and data management
    a. Can some data be omitted or truncated?
    b. Is pattern matching and post-processing validation necessary for every recording?
    c. How can more sites, species, and recordings be added without becoming unmanageable?
3. The best ways to assess presence/absence, and frequency/duration
4. What other platforms exist, such as BirdNet, and how they compare to Arbimon
    a. Would it be worth switching to a different platform?
    b. How would an inhouse tool, like a custom CNN compare, to existing platforms?

METHODOLOGY

I set up a separate Arbimon sandbox project to work with Listen Up Collab data without affecting existing projects. Using a project backup from the main admin, I focused on the

validated 2022 “All Sites” playlist (~10,300 recordings) and re-uploaded just those recordings, then rebuilt the “All Sites” playlist and needed subsets in the new project. Using Python in Jupyter Notebooks, I combined and joined the backup CSVs (recordings, sites, playlists, pattern matching results, and species) to identify the correct files by site and datetime for download and upload. I ended with site-specific datasets and playlists for a subset of Listen Up Collab sites, with a particular focus on the spotted towhee common song. This created a manageable, well-labeled subset of Listen Up Collab data that I could use to test Arbimon tools. **More detail on the data preparation steps are provided in the Appendix.**

RESULTS & OBSERVATIONS

Below are answers to the research questions, which were determined after testing the Arbimon tools on a subset of Listen Up Collab data.

Understanding the full capacity of Arbimon

If the Listen Up Collab team continues with Arbimon, the most useful tool to invest in is Random Forest. The other tools (Audio Event Detection, Clustering, and Soundscape) are helpful for exploratory analysis and broad patterns, but are less effective for presence/absence detection than Random Forest Models and Classification.

The team has already done extensive work with Pattern Matching, which can now be used to build random forest training sets, develop models, and run classifications. Once set up, random forest models can automate presence/absence classification in new recordings and dramatically reduce the manual effort required by Pattern Matching.

One limitation is that Random Forest models are restricted to one model per species sound. If a species has multiple songs, calls, or alarms, a separate model is needed for each, and results then need to be combined to understand true presence vs absence for that species at a given site and time (similar to the pattern matching tool).

How to improve data efficiency and data management

There are a few options and aspects to consider below:

Can some data be omitted or truncated?

Some data can be omitted or truncated to make processing easier. This can happen in a pre-processing step in R or Python before uploading new data to Arbimon. For example, based on my Soundscape exploration, most bird activity in summer occurs between about 5–11am and 6–9pm, so uploads could be limited to these hours to reduce data volume. Another option is to remove recordings that are mostly silence, human noise, or background noise (cars, rain).

The trade-off in both cases is the risk of missing rare events or species outside the selected times or filters. Pre-processing choices should be guided by the research questions and the balance between data reduction and potential data loss.

Is pattern matching and post-processing validation necessary for every recording?
No. Once a model is trained, such as a random forest model, pattern matching and post-processing validation are no longer needed for new recordings. Pattern matching is only required to build the initial training sets. In the case of random forest, ideally, 200–500 validated detections per species/sound are completed manually with pattern matching. These results are then used to train the model, and then pattern matching is not needed again for that species/sound.

How can more sites, species, and recordings be added without becoming unmanageable?
The Abrimon workflow I recommend below supports scaling to more sites, species, and recordings. Pre-processing can reduce unwanted recordings before the upload, and random forest models can then be built per species and per species sound. One thing to be aware of is that this can become labor-intensive for species with many calls/sounds/dialects (still substantially faster than using pattern matching alone). As an alternative, a single CNN model could be used to handle multiple call types and multiple species at once. Then adding new recordings and species would be even more manageable. More detail on this option is provided below.

The best ways to assess presence/absence, and frequency/duration
Random Forest in Arbimon is a strong option for quickly assessing presence vs absence of a specific species or call. A CNN would be even more efficient at this task. For frequency and duration, further investigation is still needed and is to be determined.

What other platforms exist, such as BirdNET, and how do they compare to Arbimon?
Within BirdNET, BirdNET-Analyzer is the most useful tool for automated, multi-species analysis and classification. Its pre-trained CNN model covers more than 6,000 species, which is powerful for broad, multi-species work. However, the out-of-the-box model may not fully match Listen Up Collab's target species or dialects. BirdNET-Analyzer is customizable, so Listen Up Collab can adapt or fine-tune the model using their own recordings and species lists. This allows the team to leverage a CNN without building one from scratch, while tailoring it to their locations and species of interest.

Arbimon vs. BirdNET-Analyzer (conceptual comparison)

Aspect	BirdNET-Analyzer	Arbimon
Primary role	Model + processing engine for ID/detection	Web platform for storage, annotation, modeling
Model type	Pre-trained global CNN classifier (EffNetB0-like)	Per-species custom detectors + tools to build them
Species coverage (out-of-box)	>6,000 bird species worldwide	None by default; models built per project/species
Multi-species at once	Yes, single pass, multi-species probabilities	Typically many separate recognizers, combined later
Locality / time prior	Built-in eBird-based species range model	Depends on user-built models and project metadata
Extensibility	Custom classifiers via train.py, embeddings	Custom per-species recognizers, occupancy models
User interface	CLI / GUI / files; dev-friendly	Web-based interface; project / team workflows

Would it be worth switching to a different platform?

BirdNET is best viewed as a complement to Arbimon, not a full replacement. Arbimon already supports Listen Up Collab's current workflow, especially around Pattern Matching and Random Forest models. BirdNET-Analyzer becomes most valuable if the team wants scalable, multi-species classification and a simpler way to handle vocal diversity. In that case, it would be worth adding BirdNET-Analyzer alongside Arbimon and gradually testing it on Arbimon-verified data, rather than fully switching right away.

A key advantage of BirdNET-Analyzer over Arbimon's Random Forest setup is how it handles vocal diversity. In Arbimon, you typically need a separate Random Forest model for each call type or context (song, alarm, contact call, etc.), then merge those outputs to estimate presence/absence and abundance for a single species. BirdNET's CNN predicts at the species level, so all vocal variants (calls, songs, dialects) can be labeled with the same species ID and handled within one model. This reduces the number of models to build and maintain, simplifies the workflow, and can provide more robust presence/absence and abundance indices for indicator species. CNNs are also ideal for handling vast amounts of data.

RECOMMENDATIONS

Arbimon Recommendation

Below is a recommended Arbimon workflow for both existing and new recordings. This approach is designed to reduce manual effort, speed up analysis, and support expansion to more sites, species, and species sounds.

Arbimon Workflow

1. **Pre-processing (optional)**
    - Reduce data before upload (e.g., keep only peak activity hours).
    - Use Soundscape outputs to refine time windows.
2. **Upload and organize recordings**
    - Upload new data.
    - Organize into playlists by site, treatment/control, time period, etc.
3. **Exploratory data analysis**
    - Audio Event Detection: prep for clustering and identify key sounds.
    - Clustering: explore call types and dialects; annotate/validate.
    - Soundscape: assess overall acoustic activity and temporal patterns.
4. **Pattern Matching (per species/sound)**
    - Manually validate a specific species sound.
    - Aim for 200-500+ Present and 200-500+ Absent recordings.
    - Create Present/Absent playlists.
5. **Random Forest training set**
    - Build a training set from the validated playlists.
    - Aim for 100+ Present, 100+ Absent recordings.
    - Use 1–10 ROIs per recording.
6. **Random Forest model**
    - Train and tune the model for target species sound.
    - Reuse the model on future data without repeating Pattern Matching.
7. **Random Forest classification and analysis**
    - Run classification on new playlists (by site, time, treatment).
    - Compare presence/absence across sites and over time.
    - Export results for higher-level analysis if needed.
    - Combine results for big picture species presence/absence.

BirdNET Recommendation

I recommend exploring BirdNET-Analyzer in parallel with Arbimon to evaluate its performance on Listen Up Collab-specific data. If a single BirdNET-Analyzer model can reliably handle all vocal variations for each target species (songs, calls, dialects) and multiple species at once, it could greatly increase processing speed and simplify presence/absence analysis. This would reduce the need to build and maintain many separate species-sound models in Arbimon.

BirdNET-Analyzer uses a CNN on spectrograms, which is one of the strongest approaches for multi-species acoustic classification. Because the core CNN and tooling already exist, Listen Up Collab can focus on customizing the model with their own recordings and species lists, rather than building and maintaining a new CNN from scratch.

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Appendix A – Project Setup and Data Preparation

- A.1 Downloading & Re-uploading Project Data
- A.2 Pattern Matching Validation Data

A.1 Downloading & Re-uploading Project Data

To not interrupt existing Listen Up Collab Arbimon projects, I created a new Arbimon project as a sandbox/testing area. Below are the steps I took to complete the set up:

1. Have main project Admin/Owner export project backup of data.
2. Download zip of project export - Since it contained nearly 400,000 records, the space needed to open each recording would be about 2T. I decided to start by extracting the data from a specific playlist in the original project, 'All Sites' from 2022. I chose this playlist because the validation process had been thorough for these recordings, and it only contains 10,300 recordings (~30 GB).
    a. The zip contains csv files from the original project of: pattern matting results, playlists, recordings, recording tags, recording validations, sites, species, and templates.
    b. It's safe to unzip the folder without downloading all the recordings. Downloading recordings is in the next step.
3. Download recordings from the project backup - This Arbimon documentation clearly documents how to download the recordings from the backup folder. I did not download all the recordings at once. Instead, I only downloaded the recordings from the 'All Sites' playlist. To do this, I used Python in Jupyter Notebooks to:
    a. Combine all 25 recording csv files into one csv file,
    b. Add 'site_id' from sites.0001.csv to the combined recordings.csv,
    c. Identify all recording ids for the playlist 'All Sites' (in file playlist_recordings.0001.csv),
    d. Sort the combined recordings.csv by the 'All Site' ids,
    e. Create a data subset csv of only 'All Site' recordings,
    f. Use this data subset csv as the new 'recordings.csv' and follow the Arbimon download recordings steps.
4. Upload the recordings by site (to prevent overlapping datetime issues) using the Arbimon desktop app.
5. Recreate the 'All Sites' playlist, and other desired subset playlists, in the Arbimon project in the browser.

A.2 Pattern Matching Validation Data

Using Python and the project backup csv files, I created several other datasets to obtain the necessary data for testing. In order to run future audio analyses, I created a dataset of the

validated data from the pattern matching processes for ‘All Sites’. I then focused further by creating validated pattern matching results for the spotted towhee common song. These steps included:

1. Load pattern_matching_rois.0001.csv in Jupyter,
2. Create a dataset of only pattern matchings in the All Sites recordings dataset,
3. Add site_id, site name, and datetime to All Sites pattern matching dataset,
4. Filter by spotted towhee, where species id = 10176. Species id found in species.0001.csv from project backup,
5. Filter further by ‘validated’ column = 1, meaning the spotted towhee has been validated as present. Save this dataset as a Present csv,
6. Filter by ‘validated column = 0, meaning the spotted towhee has been validated as not present. Save this dataset as an Absent csv,
7. Use each of these spotted towhee (SPTO) Present vs Absent csv datasets to filter through the downloaded All Sites recordings. I did this for only a few random sites from the All Sites playlist: 32159, 32160, 32210, 32563. I created new folders with the sorted recordings by Present and Absent per site, and then uploaded these to Arbimon using the desktop app. This step was important for future training sets in Random Forest and eliminating redundant pattern matching work.

Note: ‘recording_id’ in the project backup is not the same as the ‘recording_id’ once re-uploaded. Arbimon assigns a new recording_id upon upload, so files cannot be linked from the csv files and uploaded files via id. The best way to link them is by datetime and site.

Appendix B – Pattern Matching to Random Forest Training Sets

- B.1 Pattern Matching Workflow
- B.2 Random Forest Models Overview
- B.3 Steps to Build Training Set

B.1 Pattern Matching Workflow

The first step to creating Random Forest (RF) models for automated classification of species sounds is Pattern Matching. Pattern Matching allows for the identification of likely sounds based on a template, and manual review to create a confident dataset. Recordings that are marked as "present" in the Pattern Matching validation step will be automatically added to the RF training set in the next step. Recordings marked "not present" in this step is not the same as saying the sound is "absent". Labeling "absent" recordings is an extra step in the Visualizer.

Currently, "present" validations cannot be removed from the RF training set without deleting the validation in the Pattern Matching step. So, beginning Pattern Matching with a plan in mind can allow for a smoother process. (Arbimon has communicated they are working on a feature to accommodate this issue). I recommend working with one playlist at a time and validating "present" 100-200 recordings per target species/sound. This will make future steps for RF model creation manageable.

Note: You will have to manually go through recordings in the Visualizer and mark "Absent" for the absent target species sound an equal number of times as there are "present" validations. 100-200 is manageable while still producing good RF model results.

Create a Pattern Matching Template

1. Navigate to Pattern Matching → Templates → New Template
2. Name the template to reflect the species and sound type. Example: SPTO_CommonSong_Template#_Playlist#
3. In the Visualizer, find a high-quality example of the target species sound. Use the cursor to draw a tight box around the sound in the spectrogram. This box becomes your template that the algorithm will search for
4. Note**:** The quality of your template matters. Choose a clear example with minimal background noise and typical frequency/duration characteristics for the species

Run Pattern Matching on a Playlist

1. Navigate to Pattern Matching → Jobs → New Job
2. Select your template from the drop down menu

3. Under 'Recordings', select **'Playlist'** and choose a single playlist to search. Example: SPTO_2022_Site1_May
4. **Important:** Focus on one playlist at a time for thorough validation. Running pattern matching across your entire project will generate thousands of results that become unmanageable to validate and too lengthy for the RF model training sets.
5. Adjust the matching threshold if needed (lower = more sensitive, more false positives; higher = more specific, may miss true positives). Default settings are often a good starting point
6. Run the job and wait for results

Validate Pattern Matching Results

1. Once the job completes, click on the job name to view results
2. The results page shows all potential matches ranked by similarity to your template
3. Click through each result and both look at the spectrogram and listen to confirm whether it's a true match
4. If the detection contains your target species sound, mark it as 'Present' (check mark)
5. If the detection is a false positive (not the target species), mark it as 'Not Present' (x)
6. **Aim for 100-200 validated 'Present' results per playlist.** These validations are automatically added to your RF training set later.
7. **Note:** Keep your total validated 'Present' results under 500 across all playlists, because you'll need to manually assign an equal number of 'Absent' validations in the Visualizer later.

Repeat for Additional Playlists (if needed)

1. If you need more present validations, run Pattern Matching on another playlist from different sites, dates, or acoustic conditions
2. Validate those results the same way
3. Stop when you reach your target number of present validations (e.g., 150-200 total across all playlists)

Tips for Efficient Validation

- **Work manageable playlist by playlist** rather than running one massive Pattern Matching job across your entire project
- **Validate thoroughly as you go**—it's easier to validate 200 results immediately after a job runs than to come back to thousands of unvalidated detections later. Being confident in your results is essential for a well trained RF model later
- **Track your progress**: Check your training set regularly (Random Forest Models → Training Sets) to see how many present validations you've accumulated

- **Quality over quantity**: 150 high-confidence present validations are better than 3,000 questionable ones

B.2 Random Forest Models Overview

Arbimon's Random Forest model is an automatic call detector that learns what a species' sound looks like on a spectrogram and then scans long recordings for similar patterns. You first create a training set with examples of "present" and "absent" recordings, plus a clean example of the target sound (ROI), and the model then predicts for each new audio segment whether that species is calling or not.

After exploring the Random Forest model in Arbimon, I believe this tool could be a great next step for the Listen Up Collab team. Instead of repeating the pattern matching/validation process for the same species at new sites or for new recordings, the team can use their existing validation work to train random forest models, which can then classify presence/absence in new data. Training the random forest models still requires time and effort, but once the process is set up, the presence/absence classification process for new and existing data will be substantially faster and require less manual work.

Each species, and each species call/song, requires its own random forest training set and model. So if there are 16 species of interest, that's 16 random forest training sets and models. If each species has 2 sounds of interest, that's 32 training sets and models, and so on. A strong training set ideally contains 100-200 recordings of the targeted sound's presence, and 100-200 of its absence. **Keeping the training set under 500 total validations is recommended**, because absent validations must be manually assigned in the Visualizer for each present validation. Accuracy is very important, so strong confidence in presence/absence is needed. Additionally, one excellent example of the call needs to be selected as the ROI, or Region of Interest. The model will use this ROI as an example to train on.

Note: An Arbimon random forest model can technically be trained on 50+ present and 50+ absent spectrograms, but 100-200 for each is ideal for balancing model performance with manual validation workload.

B.3 Steps to Build Training Set

To create a random forest training set, present validations are automatically added from Pattern Matching, while absent validations must be manually assigned in the Visualizer. This workflow is most efficient when Pattern Matching is run on a focused playlist (rather than the entire project), with thorough validation of results as you go.

Create a New Training Set

1. Navigate to Random Forest Models → Training Sets → New Training Set
2. Name the training set and select the species sound. Example: SPTO_CommonSong_TrainingSet, Pipilo maculatus, Common Song
3. Note**:** If you create a second training set with the same species and sound type, all presence/absence validations will be added to both training sets. The only thing that can change in a new training set for the same species/call is the ROI. This can be useful for testing purposes to find the best ROI for the model.

Add Present Validations via Pattern Matching

1. Run a Pattern Matching job on a curated playlist (see Pattern Matching section for details)
2. Review the Pattern Matching results and validate detections as "Present" or "Not Present"
3. **All validations marked as "Present" are automatically added to your training set**—no additional steps needed
4. Continue validating until you have your desired number of present samples (e.g., 100-200)

Add Absent Validations via Visualizer

1. Navigate to Explore → Visualizer
2. In the top left window, select the 'Browse Recordings by Playlist' option, and select the same playlist you used for Pattern Matching
3. Click on the image of a spectrogram to make it appear in the larger window on the right, and both look and listen to confirm the species' sound is absent
4. Under 'Species Presence Validation' on the left panel, select your species and call from the drop down menu (such as Pipilo maculatus simple call). Then, under 'Birds' and your species and sound, select the Absent 'Annotation' (yellow x)
5. **Note:** You must click on each image in the top left for the correct image to appear in the right window
6. Repeat for recordings where the target species is not present until you have a balanced number of absent samples matching your present count (e.g., 100-200 absent to match 100-200 present)
7. **Tip:** If you thoroughly validated your Pattern Matching results on a playlist, you can efficiently add absences by going through that same playlist in the Visualizer and marking recordings that don't contain the target species as absent

Create an ROI (Region of Interest)

1. Find the best example of the species sound in your validated data
2. When found, navigate to that recording in the Visualizer and confirm it's marked as "Present" under 'Species Presence Validation'
3. Under 'Training Sets (RF Algorithm)', select the training set from the drop down menu (i.e., SPTO_CommonSong_2022). When selected, it will very subtly 'glow' around the edges of the selection box
4. While it's selected and 'glowing', use the cursor to draw a tight box around the target sound
5. **Alternative:** You can use an existing Template from 'Templates (Pattern Matching Analysis)' as the ROI
6. **Note:** More ROIs is not better. The model will only look at one ROI as an example in its training. Finding the best ROI is ideal.
7. **Note:** One potential limitation of this ROI method in the Arbimon Random Forest tool is that variation in the species sound may not be captured. There are more robust models outside of Arbimon that address this (i.e. BirdNET-Analyzer).

Verify Your Training Set

1. In Random Forest Models → Training Sets, click on the training set to see the tally of samples added under 'Validations', as well as the **'ROI Count'**
2. **It's important to have the same number of present and absent samples for a balanced training set** (e.g., 154 present / 154 absent)

Appendix C – Random Forest Models & Classification

- C.1 Steps to Train a Random Forest Model
- C.2 Random Forest Classification
- C.3 Steps to Run Random Forest Classification
- C.4 Findings

C.1 Steps to Train a Random Forest Model

1. To create a Random Forest model, navigate to Audio Analyses -> Random Forest Models -> New Training Job. Select model name, model (Pattern Matching with Random Forest), and the training set you created above. Then choose the training/validation breakdown. A 80/20 split is common (80% for training (fitting) and 20% in validation), but this can be experimented with. Select Create.
    a. Note: The models are pretty fast, taking only a few seconds-minutes for ~200 samples.
2. Once the model has run, you can inspect the model details by clicking on the model in 'Training Results.' You can inspect the ROI it used to train on and how well it performed. You can also set a threshold for how strict you want the model to be when it thinks it found presence/absence.
3. Experiment with new models, different thresholds, and sample sizes/ROIs until you achieve the best model results (being mindful of over and underfitting).

C.2 Random Forest Classification

Once we've trained the random forest on a set of manually reviewed recordings, it can automatically predict presence or absence of the call across thousands of new recordings without having to redo the manual pattern-matching each time. Instead of sliding a template over every spectrogram, the model has learned what the call looks like and quickly flags likely detections in the full dataset, so the manual work is limited to training and spot-checking rather than reviewing every file.

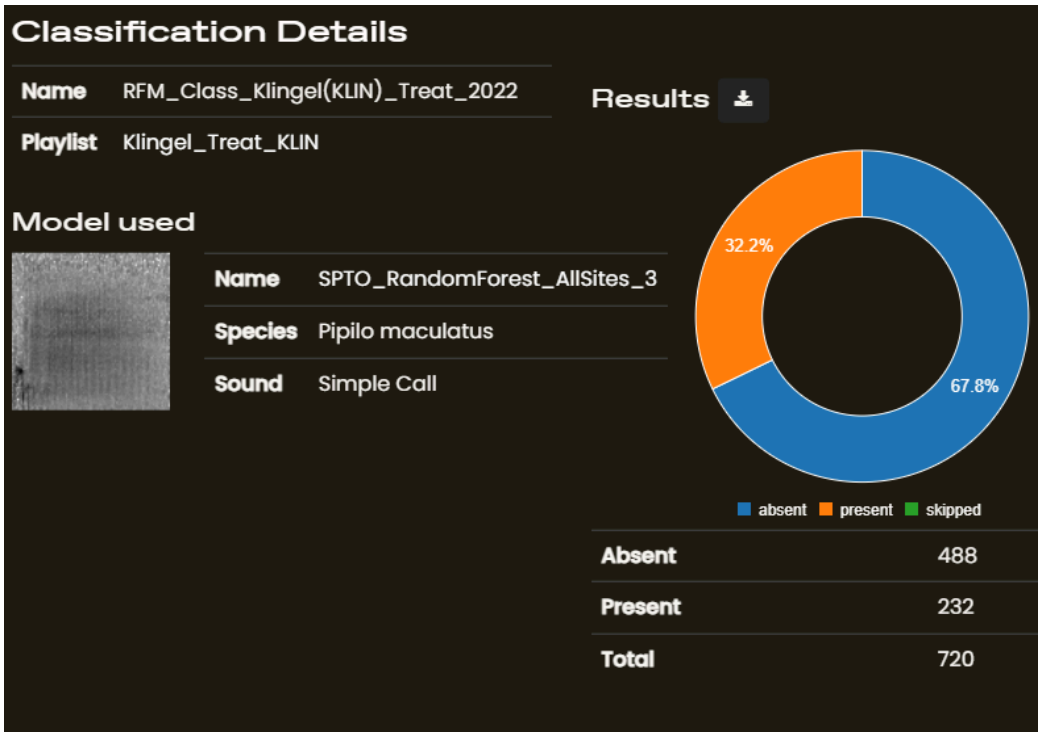
C.3 Steps to Run Random Forest Classification

1. To classify data based on the target species and sound, navigate to Random Forest Models -> New Classification Job. Create a name, select the model you trained in the previous steps (the best model achieved), and select a playlist of the recordings to classify. Then select Create.
2. Progress can be seen in Audio Analyses -> Active Jobs.
    a. Note: You may need to refresh the pages to see the progress change.
    b. Short playlists of about 500 records takes about 10-20 minutes to run. Playlists of 10,000 records might take a couple hours.

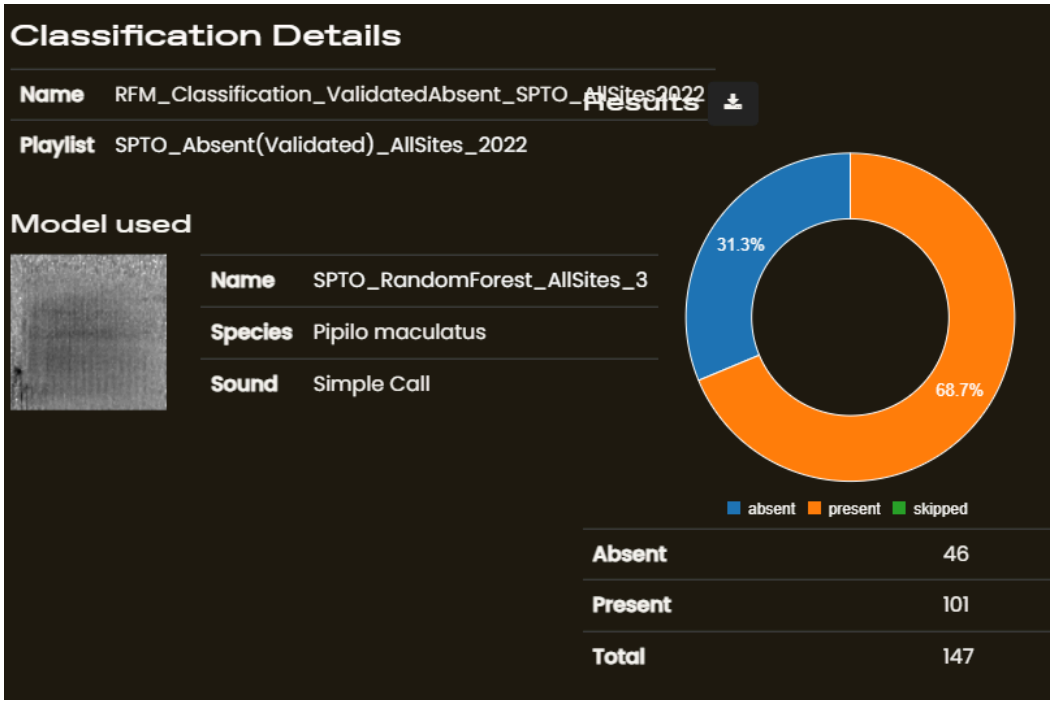
3. View the results in Random Forest Models -> Classification Results, by selecting the classification you ran. It gives a quick overview of the image and model used in run, as well as a numeric and visual breakdown of the presence vs absence results. Selecting 'Show details per recording' provides more details and an ability to more closely inspect model accuracy by checking the spectrograms.

C.4 Findings

I created playlists for each site of interest, and then ran a random forest classification on each of those playlists. I experimented with a few sites from the All Sites 2022 data, and ran both the control and treatment playlists for each of those sites. It was important to have the same number of records in each playlist for the same sites in order to have a balanced results comparison. From the quick runs and visual displays, I was quickly able to encounter some surprising (to me) findings, that the spotted towhee common song is more abundant in treatment sites than control sites, across multiple locations in the 2022 data.



I also ran the validated Absent and validated Present playlists through the classification model, and confirmed what I had seen from manual inspection. In this particular case, the validated Present playlist contains about 20% recordings without clear SPTO presence, and the validated Absent playlist contains nearly 70% data considered 'present'. This confirmed my belief that the extra step of validating presence/absence during the training set creation, from already validated records, can be an important extra step in the pipeline. The better the Pattern Matching results, the faster and more robust the training set creation step will be for Random Forest models.

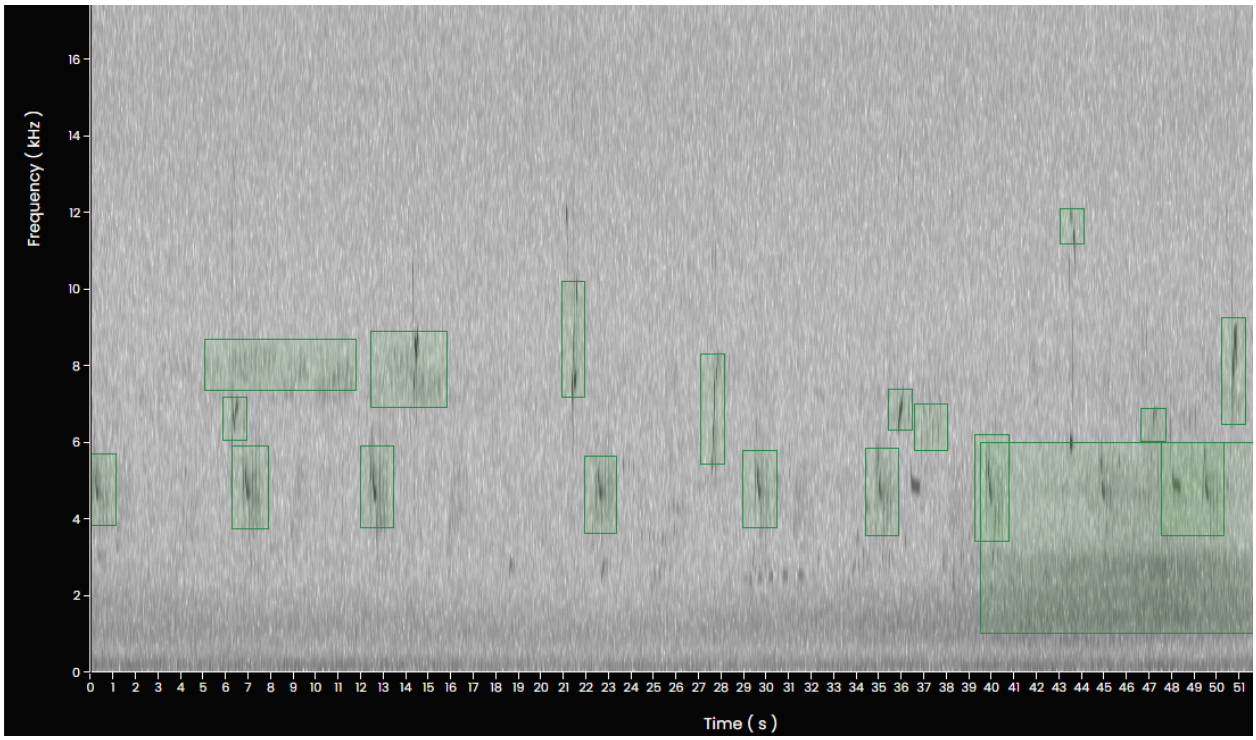

Classification Details
Name
RFM_Classification_ValidatedAbsent_SPTO_AllSites2022
Playlist
SPTO_Absent(Validated)_AllSites_2022
Model used
Name
SPTO_RandomForest_AllSites_3
Species
Pipilo maculatus
Sound
Simple Call
Results
31.3%
68.7%
absent
present
skipped
Absent
46
Present
101
Total
147

Appendix D - Audio Event Detection, Clustering, & Soundscapes

- D.1 Audio Event Detection Overview
- D.2 Clustering Overview
- D.3 Soundscape Overview

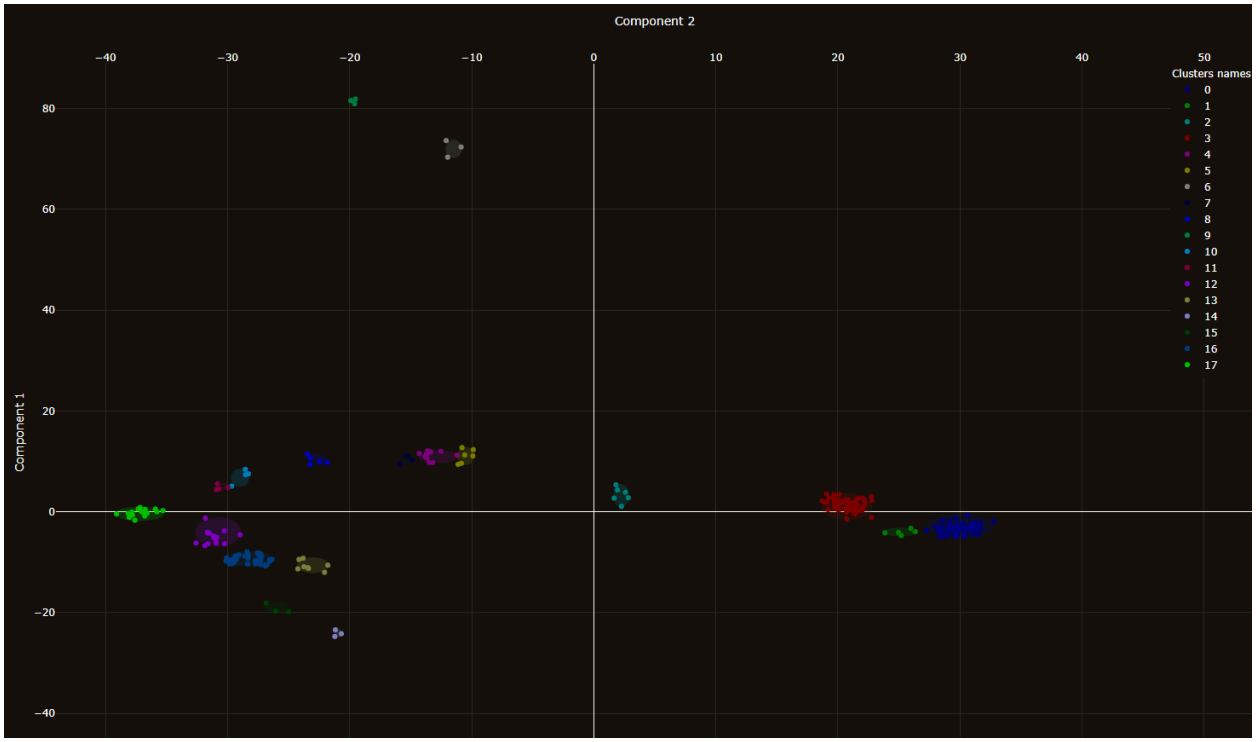
D.1 Audio Event Detection Overview

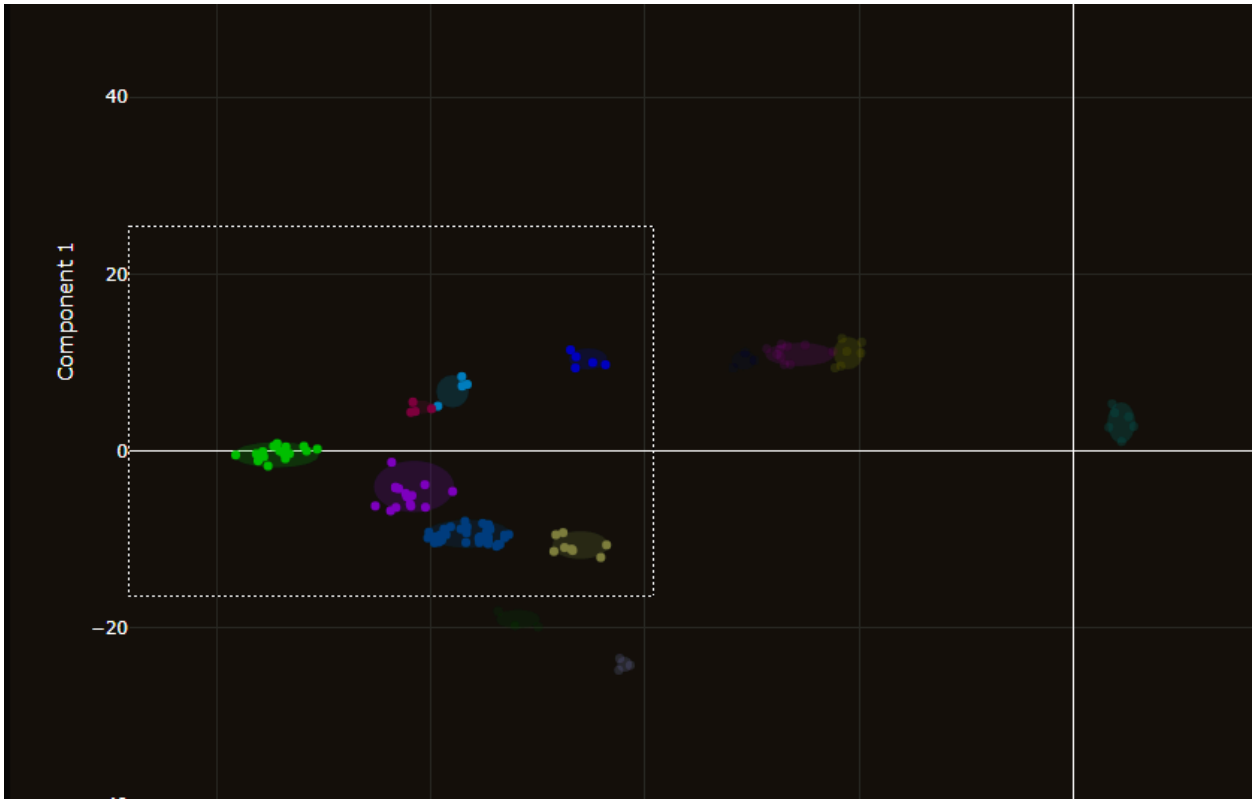
The Audio Event Detection tool automatically scans recordings and marks segments that contain any kind of sound event above background noise, or based on the parameters you set. Parameters can be broad to capture a wide variety of sounds, or narrow to capture sounds of interest. The tool highlights where and when these events occur, creating “candidates” that may include target species calls, other animals, or non-biological noise. This lets you quickly find, review, and label potential calls of interest without having to listen to every minute of audio manually. These detected sound events can then be passed into the clustering algorithm, which groups together similar events based on their acoustic characteristics.

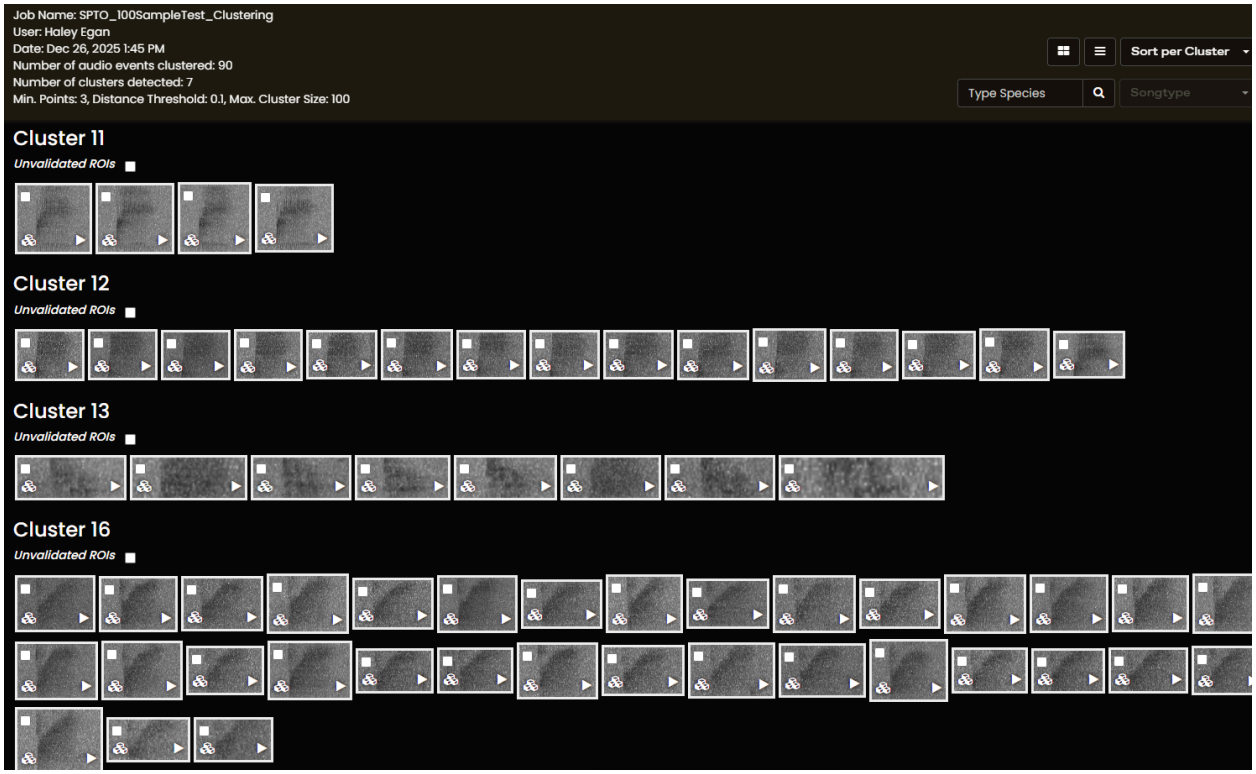


D.2 Clustering Overview

The clustering tool takes the individual sound events found by Audio Event Detection and automatically groups together those that are acoustically similar. Each cluster tends to represent a recurring sound type (e.g., a particular species call). This helps you quickly discover patterns in your data, find repeated call types, and prioritize which groups to inspect and label, instead of reviewing every event one by one.

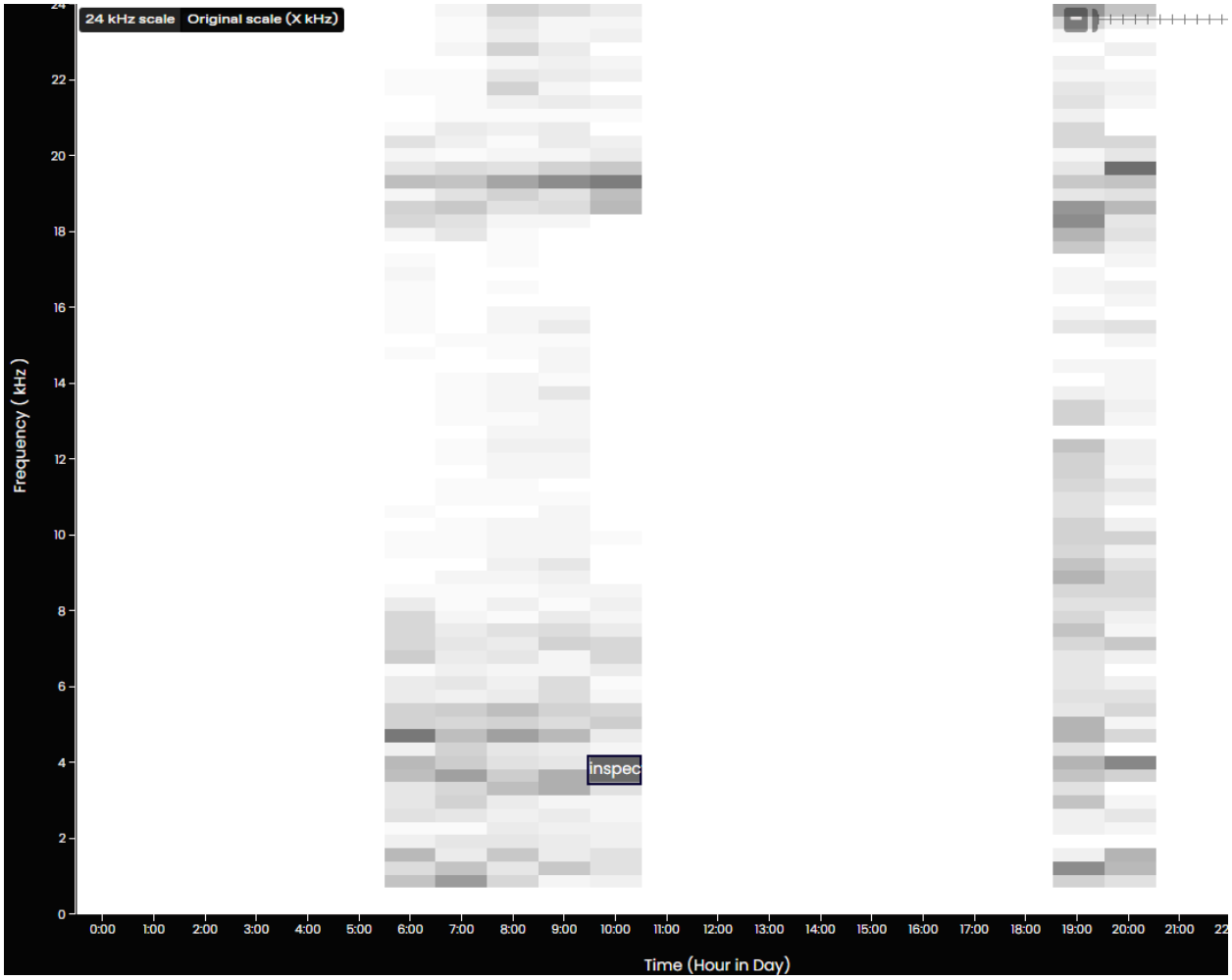

Component 2
Component 1
Clusters names
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17


Component 1



D.3 Soundscape Overview

The Soundscape tool summarizes how different sounds are distributed over time so you can see the overall acoustic “profile” of a site. It aggregates detections or acoustic indices into visual summaries (like daily or seasonal patterns), showing when biological activity or specific sound types are most common. This helps you compare sites, track changes over time, and identify broad ecological patterns without inspecting individual recordings.


24 kHz scale
Original scale (X kHz)
Frequency (kHz)
Time (Hour in Day)
inspec